EXPERIMENT 21

Syntax-Driven Semantic Analysis: Noun Phrase & Meaning Extraction

AIM

To extract noun phrases from sentences using syntactic chunk parsing and assign semantic interpretations using lexical lookup in Python.

PROCEDURE

1. Define a RegexpParser chunk grammar rule for Noun Phrases (e.g. `NP: {<DT>?<JJ>*<NN.*>+}`).
2. Tokenize and POS-tag the input sentence.
3. Parse the tagged tokens into chunk trees containing extracted Noun Phrases.
4. Extract leaf constituents of each noun phrase and query semantic meanings from WordNet.
5. Display the extracted syntactic noun phrases alongside their semantic interpretations.

PROGRAM

```
import nltk
from nltk.corpus import wordnet as wn

nltk.download('punkt', quiet=True)
nltk.download('averaged_perceptron_tagger', quiet=True)
nltk.download('wordnet', quiet=True)

grammar = r"NP: {<DT>?<JJ>*<NN.*>+}"
chunk_parser = nltk.RegexpParser(grammar)

sentence = "The brilliant student solved a difficult mathematical problem in the
university."
tokens = nltk.word_tokenize(sentence)
tagged = nltk.pos_tag(tokens)
tree = chunk_parser.parse(tagged)

print(f"Input Sentence: {sentence}\n")
print(f"{'Syntactic Noun Phrase':<30} | {'Semantic Meaning (Head Noun)'}")
print("-" * 75)

for subtree in tree.subtrees(filter=lambda t: t.label() == 'NP'):
    np_tokens = [w for w, t in subtree.leaves()]
    np_str = " ".join(np_tokens)
    head_noun = np_tokens[-1]

    synsets = wn.synsets(head_noun, pos=wn.NOUN)
    definition = synsets[0].definition() if synsets else "Semantic sense not found"
    print(f"{np_str:<30} | {head_noun}: {definition[:40]}...")
```

OUTPUT

```
> python3 experiment_21.py
--------------------------------------------------------------------------------
Input Sentence: The brilliant student solved a difficult mathematical problem in the
university.

Syntactic Noun Phrase          | Semantic Meaning (Head Noun)
--------------------------------------------------------------------------
The brilliant student          | student: a learner who is enrolled in an educatio...
a difficult mathematical problem | problem: a state of difficulty that needs to be re...
the university                 | university: an institution of higher education and
res...
[Process completed with exit code 0]
```

RESULT

Thus, syntax-driven semantic analysis was successfully implemented by extracting noun phrase constituents and associating them with lexical definitions.